

MPSUB: A MOMENTUM P -DIMENSIONAL SUBSPACE TRUST-REGION METHOD FOR DERIVATIVE-FREE FINE-TUNING OF LARGE LANGUAGE MODELS*

YUYANG WANG[†], HAOYU YAO[‡], AND PENGCHENG XIE[§]

Abstract. Full-parameter fine-tuning of large language models has a substantial memory cost because backpropagation requires storing activations and gradients. Zeroth-order optimization avoids that by estimating update directions from loss evaluations, but existing methods often depend on a learning rate that must be retuned for each model and task. We propose the momentum p -dimensional subspace trust-region method (MpSub). At each iteration, MpSub searches within a p -dimensional subspace: one direction preserves the historical information carried by the most recent accepted step, while the remaining directions promote exploration through fresh random sampling. The subspace gradient is estimated by central differences, a trial step is computed from a linear trust-region model, and the trust-region radius is updated according to the agreement between predicted and observed loss reduction. The radius controls both the finite-difference perturbation and the trust-region bound on the subspace step, so no learning rate is required. In the LLM setting, all evaluations within an iteration share one minibatch, and the directions are regenerated on the fly and applied in place, so training uses forward evaluations alone. For smooth deterministic objectives under Gaussian direction sampling, we bound the finite-difference error, quantify the gradient captured by the random subspace, and establish almost-sure global convergence ($\liminf_{k \rightarrow \infty} \|\nabla f(\mathbf{x}_k)\|_2 = 0$) under a safeguarded radius update. Under a matched budget of 8,400 training-objective forward passes on CommitmentBank, MpSub achieves mean test accuracies of 0.673 on OPT-125M and 0.690 on OPT-350M using fixed parameters, matching the performance of MeZO with development-selected learning rates (0.685 at both sizes).

Key words. derivative-free optimization, subspace trust region, zeroth-order optimization, large language model fine-tuning

AMS subject classifications. 90C56, 65K05, 90C26

1. Introduction. Zeroth-order (ZO) optimization has become an attractive alternative for full-parameter fine-tuning of large language models (LLMs) when backpropagation is limited by memory. Instead of storing activations for a backward pass, ZO methods estimate update directions from function evaluations. MeZO [11] applies the classical two-point random estimator [13] to language models and regenerates random perturbations from seeds rather than storing them. Later methods have modified the perturbation structure, including random low-dimensional subspaces [15] and low-rank gradient structures [5].

These methods still use a prescribed learning rate to determine the update size. Its effective value can vary substantially across settings and may require a separate search. In addition, a single random direction contains only a small amount of directional information in a very high-dimensional parameter space. For a normalized isotropic direction, the typical magnitude of its projection onto a fixed gradient decreases as the dimension grows. This motivates using more than one search direction while avoiding a proportional increase in stored parameter-sized vectors.

Trust-region methods provide a natural way to control the step size without prescribing a learning rate. In derivative-free optimization, the trust-region radius is adjusted according to the agreement between a local model and the observed objec-

*Preprint.

[†]Xi'an Jiaotong University, Xi'an, China (vibrancy05@stu.xjtu.edu.cn).

[‡]Xi'an Jiaotong University, Xi'an, China (yaohaoyu@stu.xjtu.edu.cn).

[§]Lawrence Berkeley National Laboratory, Berkeley, CA, USA (pxie@lbl.gov). Corresponding author.

tive values [1, 8, 6, 7]. Direct model construction in the full parameter space is not practical for models with hundreds of millions of variables, which motivates restricting the search to a much smaller subspace [4, 16, 17]. Randomized subspaces have also been studied in derivative-free optimization [3, 10, 12].

The closest method to the present work is 2D-MoSub [14]. It searches in a two-dimensional subspace formed by the previous accepted step and a new search direction. The previous step carries information from the optimization trajectory, while the new direction provides exploration. A quadratic interpolation model is constructed in this two-dimensional subspace and minimized within a trust region. This momentum-plus-exploration construction is also the starting point of our method. The difficulty in extending the same approach to a larger subspace lies in the quadratic model: a direct extension to dimension p would require $\frac{1}{2}(p+1)(p+2)$ interpolation points. At $p = 20$, this would require 231 points before a trial step is computed.

We therefore consider the LLM fine-tuning problem

$$(1.1) \quad \min_{\mathbf{x} \in \mathbb{R}^n} \mathbb{E}_{\xi \sim \mathcal{D}} [\ell(\mathbf{x}; \xi)],$$

and propose **MpSub**, a momentum p -dimensional subspace trust-region method. **MpSub** keeps the momentum-plus-exploration structure of 2D-MoSub but replaces quadratic interpolation with directional derivatives estimated by central differences. These estimates define a linear model in a p -dimensional subspace formed by the most recent accepted displacement and $p - 1$ fresh Gaussian directions. The resulting iteration requires $2p + 2$ function evaluations, so the evaluation cost grows linearly with the subspace dimension. For LLM fine-tuning, all evaluations within one iteration use the same minibatch, and the random directions are regenerated from seeds when needed rather than stored explicitly.

For smooth deterministic objectives, we analyze the Gaussian directions directly without orthogonality assumptions. We derive finite-difference error bounds, quantify the gradient captured by the random subspace, and prove almost-sure global convergence ($\liminf_{k \rightarrow \infty} \|\nabla f(\mathbf{x}_k)\|_2 = 0$) under a safeguarded trust-region radius update. We then compare **MpSub** with MeZO on OPT-125M and OPT-350M under matched forward-evaluation budgets.

2. MpSub. We consider the unconstrained optimization problem

$$(2.1) \quad \min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}),$$

in the regime where the parameter dimension n is very large (e.g., $n \sim 10^8$ in full-parameter language model fine-tuning) and derivative information is inaccessible. In our numerical experiments (Section 4), f represents the stochastic minibatch loss (1.1) resampled at each iteration, whereas in the theoretical analysis (Section 3), f is treated as a deterministic, continuously differentiable objective.

At each iteration k , **MpSub** maintains the parameter iterate $\mathbf{x}_k \in \mathbb{R}^n$, a trust-region radius $\Delta_k \in [\Delta_{\min}, \Delta_{\max}]$, and a momentum displacement $\mathbf{m}_k \in \mathbb{R}^n$ from the most recent accepted step. The radius Δ_k serves as both the finite-difference perturbation scale and the trust-region bound. The overall procedure is summarized in Algorithm 2.1.

2.1. Subspace construction. At iteration k , optimization is restricted to the affine subspace $\mathbf{x}_k + \mathcal{S}_k$, where $\mathcal{S}_k \subset \mathbb{R}^n$ is spanned by p direction vectors $\mathbf{d}_1^{(k)}, \dots, \mathbf{d}_p^{(k)}$. Gathering these directions as columns of the frame matrix $\mathbf{D}_k = [\mathbf{d}_1^{(k)}, \dots, \mathbf{d}_p^{(k)}] \in$

Algorithm 2.1 MpSub: momentum p -dimensional subspace trust-region method

```

1: Input. Initial iterate  $\mathbf{x}_0 \in \mathbb{R}^n$ , subspace dimension  $p$ , initial radius  $\Delta_0 \in [\Delta_{\min}, \Delta_{\max}]$ , contraction factor  $\gamma_1 \in (0, 1)$ , expansion factor  $\gamma_2 > 1$ , bounds  $0 < \Delta_{\min} < \Delta_{\max}$ , and threshold  $\eta \in (0, 1)$ .
2: Initialize momentum vector  $\mathbf{m}_0 = \mathbf{0}$ .
3: for  $k = 0, 1, 2, \dots$  do
4:   Step 1: Subspace frame construction.
5:   If  $\|\mathbf{m}_k\|_2 > 0$ : set  $\mathbf{d}_1^{(k)} = \mathbf{m}_k / \|\mathbf{m}_k\|_2$ ; else draw  $\mathbf{z}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n)$  and set  $\mathbf{d}_1^{(k)} = \mathbf{z}_1 / \sqrt{n}$ .
6:   For  $i = 2, \dots, p$ : draw  $\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n)$  independently and set  $\mathbf{d}_i^{(k)} = \mathbf{z}_i / \sqrt{n}$ .
7:   Form the frame matrix  $\mathbf{D}_k = [\mathbf{d}_1^{(k)}, \dots, \mathbf{d}_p^{(k)}] \in \mathbb{R}^{n \times p}$ .
8:   Step 2: Subspace gradient estimation.
9:   Evaluate  $f_0 = f(\mathbf{x}_k)$ .
10:  For  $i = 1, \dots, p$ : evaluate  $f_i^\pm = f(\mathbf{x}_k \pm \Delta_k \mathbf{d}_i^{(k)})$  and set  $g_{k,i} = (f_i^+ - f_i^-) / (2\Delta_k)$ .
11:  Assemble the subspace gradient  $\mathbf{g}_k = (g_{k,1}, \dots, g_{k,p})^\top$ .
12:  If  $\|\mathbf{g}_k\|_2 = 0$ : set  $\Delta_{k+1} = \max(\gamma_1 \Delta_k, \Delta_{\min})$ , retain  $\mathbf{x}_k$  and  $\mathbf{m}_k$ , and continue.

13:  Step 3: Trust-region trial step.
14:  Compute  $\mathbf{s}_s = -\Delta_k \mathbf{g}_k / \|\mathbf{g}_k\|_2$  and trial iterate  $\mathbf{x}^+ = \mathbf{x}_k + \mathbf{D}_k \mathbf{s}_s$ .
15:  Evaluate  $f^+ = f(\mathbf{x}^+)$ , predicted reduction  $\text{pred}_k = \Delta_k \|\mathbf{g}_k\|_2$ , and ratio  $\rho_k = (f_0 - f^+) / \text{pred}_k$ .
16:  Step 4a: Iterate acceptance.
17:  if  $f^+ < f_0$  then
18:     $\mathbf{x}_{k+1} = \mathbf{x}^+$ ,  $\mathbf{m}_{k+1} = \mathbf{x}^+ - \mathbf{x}_k$ . (accepted)
19:  else
20:     $\mathbf{x}_{k+1} = \mathbf{x}_k$ ,  $\mathbf{m}_{k+1} = \mathbf{m}_k$ . (rejected)
21:  end if
22:  Step 4b: Trust-region radius adaptation.
23:  if  $\rho_k \geq \eta$  then
24:     $\Delta_{k+1} = \min(\gamma_2 \Delta_k, \Delta_{\max})$ . (successful)
25:  else
26:     $\Delta_{k+1} = \max(\gamma_1 \Delta_k, \Delta_{\min})$ . (unsuccessful)
27:  end if
28: end for

```

$\mathbb{R}^{n \times p}$, candidate displacements within $\mathcal{S}_k$ are parameterized by a coordinate vector $\mathbf{s}_s \in \mathbb{R}^p$ via

$$(2.2) \quad \mathbf{x}(\mathbf{s}_s) = \mathbf{x}_k + \mathbf{D}_k \mathbf{s}_s = \mathbf{x}_k + \sum_{i=1}^p s_{s,i} \mathbf{d}_i^{(k)}.$$

The first search direction $\mathbf{d}_1^{(k)}$ incorporates momentum from the optimization trajectory. When an accepted step is available, it is set to the normalized displacement

$$(2.3) \quad \mathbf{d}_1^{(k)} = \frac{\mathbf{m}_k}{\|\mathbf{m}_k\|_2},$$

where $\mathbf{m}_k$ denotes the parameter change resulting from the most recent successful

step. Because consecutive descent steps in continuous optimization often exhibit directional correlation, retaining (2.3) injects historical progress into the subspace without requiring additional function evaluations. The remaining $p - 1$ directions are drawn independently as

$$(2.4) \quad \mathbf{d}_i^{(k)} = \frac{\mathbf{z}_i}{\sqrt{n}}, \quad \mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n), \quad i = 2, \dots, p,$$

with the same distribution supplying $\mathbf{d}_1^{(k)}$ when no accepted step is available. The factor $1/\sqrt{n}$ normalizes the expected Euclidean norm, $\mathbb{E}\|\mathbf{d}_i^{(k)}\|_2^2 = 1$.

A fundamental geometric property of high-dimensional Gaussian vectors is their near-orthogonality: for $i \neq j$,

$$(2.5) \quad \mathbb{E}[\mathbf{d}_i^{(k)\top} \mathbf{d}_j^{(k)}] = 0 \quad \text{and} \quad \mathbb{E}[(\mathbf{d}_i^{(k)\top} \mathbf{d}_j^{(k)})^2] = \frac{1}{n}.$$

Consequently, the pairwise inner products have root-mean-square magnitude $n^{-1/2} \approx 10^{-4}$ at $n \sim 10^8$. Since high-dimensional Gaussians are nearly orthogonal, we omit Gram–Schmidt orthogonalization to save computation and memory; the convergence of MpSub with unorthogonalized directions is established in Section 3. The dimension p controls how many directional derivatives enter the subspace model; its choice is examined empirically in Section 4.

2.2. Subspace model and trial step. For each direction $\mathbf{d}_i^{(k)}$, MpSub estimates the corresponding directional derivative via central differences:

$$(2.6) \quad g_{k,i} = \frac{f(\mathbf{x}_k + \Delta_k \mathbf{d}_i^{(k)}) - f(\mathbf{x}_k - \Delta_k \mathbf{d}_i^{(k)})}{2\Delta_k}, \quad i = 1, \dots, p.$$

Stacking these estimates yields the subspace gradient $\mathbf{g}_k = (g_{k,1}, \dots, g_{k,p})^\top \in \mathbb{R}^p$, where each entry $g_{k,i}$ approximates $\nabla f(\mathbf{x}_k)^\top \mathbf{d}_i^{(k)}$. Setting the finite-difference perturbation size equal to the trust-region radius Δ_k ties the sampling scale directly to the local trust-region size. Under the Lipschitz continuity of ∇f , the finite-difference error is bounded linearly by Δ_k , with explicit bounds established in Lemma 3.2.

The estimated gradient defines a local linear surrogate model on the subspace coordinates:

$$(2.7) \quad m_k(\mathbf{s}_s) = f(\mathbf{x}_k) + \mathbf{g}_k^\top \mathbf{s}_s.$$

The trial step is determined by minimizing this linear model subject to the trust-region constraint:

$$(2.8) \quad \min_{\mathbf{s}_s \in \mathbb{R}^p} \mathbf{g}_k^\top \mathbf{s}_s \quad \text{subject to} \quad \|\mathbf{s}_s\|_2 \leq \Delta_k.$$

Whenever $\mathbf{g}_k \neq \mathbf{0}$, problem (2.8) admits the unique closed-form solution

$$(2.9) \quad \mathbf{s}_s = -\Delta_k \frac{\mathbf{g}_k}{\|\mathbf{g}_k\|_2},$$

which corresponds to the steepest descent direction on the trust-region boundary. Mapping $\mathbf{s}_s$ back to the ambient parameter space via (2.2) yields the candidate iterate

$$(2.10) \quad \mathbf{x}^+ = \mathbf{x}_k + \mathbf{D}_k \mathbf{s}_s = \mathbf{x}_k - \frac{\Delta_k}{\|\mathbf{g}_k\|_2} \sum_{i=1}^p g_{k,i} \mathbf{d}_i^{(k)},$$

with the predicted objective reduction given by

$$(2.11) \quad \text{pred}_k = m_k(\mathbf{0}) - m_k(\mathbf{s}_s) = -\mathbf{g}_k^\top \mathbf{s}_s = \Delta_k \|\mathbf{g}_k\|_2 > 0.$$

If $\mathbf{g}_k = \mathbf{0}$, the estimated gradient vanishes across the subspace; in this event, the method contracts the trust-region radius, sets $\mathbf{x}_{k+1} = \mathbf{x}_k$ and $\mathbf{m}_{k+1} = \mathbf{m}_k$, and proceeds to the next iteration without evaluating an additional trial point.

2.3. Acceptance and radius update. Upon evaluating the candidate loss $f^+ = f(\mathbf{x}^+)$, MpSub assesses the agreement between the surrogate model and the objective through the ratio

$$(2.12) \quad \rho_k = \frac{f(\mathbf{x}_k) - f(\mathbf{x}^+)}{\Delta_k \|\mathbf{g}_k\|_2}.$$

The algorithm decouples iterate acceptance (Step 4a) from trust-region radius adaptation (Step 4b). The candidate point is accepted whenever it yields a strict objective decrease:

$$(2.13) \quad \mathbf{x}_{k+1} = \begin{cases} \mathbf{x}^+, & \text{if } f(\mathbf{x}^+) < f(\mathbf{x}_k), \\ \mathbf{x}_k, & \text{otherwise,} \end{cases} \quad \mathbf{m}_{k+1} = \begin{cases} \mathbf{x}^+ - \mathbf{x}_k, & \text{if } f(\mathbf{x}^+) < f(\mathbf{x}_k), \\ \mathbf{m}_k, & \text{otherwise,} \end{cases}$$

while the trust-region radius is updated according to the ratio test:

$$(2.14) \quad \Delta_{k+1} = \begin{cases} \min\{\gamma_2 \Delta_k, \Delta_{\max}\}, & \text{if } \rho_k \geq \eta, \\ \max\{\gamma_1 \Delta_k, \Delta_{\min}\}, & \text{if } \rho_k < \eta. \end{cases}$$

Unlike classical trust-region methods [6] that accept trial points only when $\rho_k \geq \eta$, the rules (2.13)–(2.14) separate acceptance from radius adjustment. In zeroth-order settings where forward passes dominate execution time, enforcing $\rho_k \geq \eta$ for step acceptance would discard trial points that achieve objective reduction whenever $0 < f(\mathbf{x}_k) - f(\mathbf{x}^+) < \eta \text{pred}_k$. Under (2.13), every candidate yielding a strictly lower observed loss is retained. Concurrently, the ratio ρ_k regulates the exploration scale: the trust region expands when the linear surrogate reliably predicts the objective reduction ($\rho_k \geq \eta$), and contracts when nonlinearities or estimation errors dominate ($\rho_k < \eta$).

2.4. Algorithmic realization for LLM fine-tuning. While Algorithm 2.1 is stated for a generic objective f , applying it to full-parameter fine-tuning of large language models requires addressing two practical issues: minibatch stochasticity and the memory cost of storing p search directions.

2.4.1. Consistent minibatch evaluation. In stochastic language model fine-tuning, the objective available at training step k is defined by a sampled minibatch $\mathcal{B}_k \subset \mathcal{D}$:

$$(2.15) \quad f_k(\mathbf{x}) = \frac{1}{|\mathcal{B}_k|} \sum_{\xi \in \mathcal{B}_k} \ell(\mathbf{x}; \xi).$$

To prevent batch sampling noise from distorting the trust-region ratio, the minibatch $\mathcal{B}_k$ is held fixed throughout iteration k : the baseline loss $f_k(\mathbf{x}_k)$, the $2p$ central-difference evaluations $f_k(\mathbf{x}_k \pm \Delta_k \mathbf{d}_i^{(k)})$, and the trial evaluation $f_k(\mathbf{x}^+)$ are all computed on the same minibatch $\mathcal{B}_k$. Holding $\mathcal{B}_k$ constant ensures that the estimated

subspace gradient $\mathbf{g}_k$ and the agreement ratio

$$(2.16) \quad \rho_k = \frac{f_k(\mathbf{x}_k) - f_k(\mathbf{x}^+)}{\Delta_k \|\mathbf{g}_k\|_2}$$

reflect local model fidelity on a fixed sample rather than minibatch variance.

Across iterations, a fresh minibatch $\mathcal{B}_{k+1}$ is drawn, while Δ_k and $\mathbf{m}_k$ carry over to the next step. Each iteration expends $2p + 2$ forward passes on the training objective (42 passes when $p = 20$).

2.4.2. In-place perturbations and seed regeneration. Storing the frame matrix $\mathbf{D}_k \in \mathbb{R}^{n \times p}$ explicitly would require pn floating-point numbers. For an LLM such as OPT-125M ($n \approx 1.25 \times 10^8$) with $p = 20$ in FP32, this would require approximately 10 GB of memory, which exceeds the parameter storage of the model itself.

To avoid this memory overhead, MpSub never materializes the matrix $\mathbf{D}_k$ in memory. Instead, each exploratory direction $\mathbf{d}_i^{(k)}$ ($i = 2, \dots, p$) is generated on the fly using a deterministic seed $\text{seed}(k, i) = (\text{base_seed} \cdot C_1 + k \cdot C_2 + i \cdot C_3) \bmod M$, where base_seed is the initial random seed of the training run, C_1, C_2, C_3 are fixed primes, and $M = 2^{31} - 1$. Weights are perturbed, evaluated, and restored in place, reconstructing directions as needed without storing them.

Once the subspace gradient $\mathbf{g}_k$ and the coordinate step $\mathbf{s}_s = -\Delta_k \mathbf{g}_k / \|\mathbf{g}_k\|_2$ are computed, the directions are regenerated a final time and combined in place to form the trial displacement

$$(2.17) \quad \mathbf{u}_k = \sum_{i=1}^p s_{s,i} \mathbf{d}_i^{(k)}.$$

The displacement $\mathbf{u}_k$ is accumulated in an auxiliary buffer of dimension n while advancing the model weights to $\mathbf{x}^+ = \mathbf{x}_k + \mathbf{u}_k$. If the trial step is accepted, $\mathbf{u}_k$ is preserved as the new momentum displacement $\mathbf{m}_{k+1}$; if rejected, subtracting $\mathbf{u}_k$ immediately restores the base weights $\mathbf{x}_k$.

Training is thus executed through forward inference passes alone. Auxiliary storage is confined to the momentum vector $\mathbf{m}_k$ and a single trial-step buffer of size n , keeping the training memory footprint at the inference level and independent of the subspace dimension p .

3. Theoretical Analysis. We analyze MpSub for a deterministic objective under the Gaussian direction sampling used in Algorithm 2.1. No orthogonality between the search directions is assumed. The analysis first gives error and descent bounds for a general frame $\mathbf{D}_k$, and then uses the distribution of the Gaussian directions to obtain a global first-order convergence result.

Throughout this section, we set $\Delta_{\min} = 0$ for the asymptotic analysis. We also use the following safeguarded radius update with a fixed constant $\kappa > 0$:

$$(3.1) \quad \Delta_{k+1} = \begin{cases} \min\{\gamma_2 \Delta_k, \Delta_{\max}\}, & \text{if } \rho_k \geq \eta \text{ and } \|\mathbf{g}_k\|_2 \geq \kappa \Delta_k, \\ \gamma_1 \Delta_k, & \text{otherwise.} \end{cases}$$

As in Algorithm 2.1, a trial step is accepted whenever $f(\mathbf{x}^+) < f(\mathbf{x}_k)$.

ASSUMPTION 3.1. *The objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuously differentiable and bounded below. Its gradient is Lipschitz continuous with constant $L > 0$:*

$$(3.2) \quad \|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\|_2 \leq L \|\mathbf{x} - \mathbf{y}\|_2, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

Let $\mathbf{D}_k = [\mathbf{d}_1^{(k)}, \dots, \mathbf{d}_p^{(k)}] \in \mathbb{R}^{n \times p}$ be the frame generated at iteration k , and define the exact directional derivatives

$$\hat{g}_{k,i} := \nabla f(\mathbf{x}_k)^\top \mathbf{d}_i^{(k)}, \quad \hat{\mathbf{g}}_k := \mathbf{D}_k^\top \nabla f(\mathbf{x}_k).$$

The finite-difference estimates are denoted by

$$\mathbf{g}_k = (g_{k,1}, \dots, g_{k,p})^\top, \quad g_{k,i} = \frac{f(\mathbf{x}_k + \Delta_k \mathbf{d}_i^{(k)}) - f(\mathbf{x}_k - \Delta_k \mathbf{d}_i^{(k)})}{2\Delta_k}.$$

3.1. Finite differences and random subspaces. We first bound the finite-difference error without assuming that the directions have unit norm.

LEMMA 3.2 (Finite-difference error). *Suppose Assumption 3.1 holds. Then*

$$(3.3) \quad |g_{k,i} - \hat{g}_{k,i}| \leq \frac{L\Delta_k}{2} \|\mathbf{d}_i^{(k)}\|_2^2, \quad i = 1, \dots, p.$$

Consequently,

$$(3.4) \quad \|\mathbf{g}_k - \hat{\mathbf{g}}_k\|_2 \leq \frac{L\Delta_k}{2} Q_k, \quad Q_k := \left(\sum_{i=1}^p \|\mathbf{d}_i^{(k)}\|_2^4 \right)^{1/2}.$$

Proof. For any direction $\mathbf{d}$ and $t > 0$, Lipschitz continuity of the gradient gives

$$|f(\mathbf{x}_k + t\mathbf{d}) - f(\mathbf{x}_k) - t\nabla f(\mathbf{x}_k)^\top \mathbf{d}| \leq \frac{L}{2} t^2 \|\mathbf{d}\|_2^2.$$

Applying this bound to $t = \Delta_k$ in the directions $\mathbf{d}_i^{(k)}$ and $-\mathbf{d}_i^{(k)}$, write

$$f(\mathbf{x}_k + \Delta_k \mathbf{d}_i^{(k)}) = f(\mathbf{x}_k) + \Delta_k \hat{g}_{k,i} + R_{i,+},$$

and

$$f(\mathbf{x}_k - \Delta_k \mathbf{d}_i^{(k)}) = f(\mathbf{x}_k) - \Delta_k \hat{g}_{k,i} + R_{i,-},$$

where $|R_{i,\pm}| \leq \frac{L\Delta_k^2}{2} \|\mathbf{d}_i^{(k)}\|_2^2$. Subtracting the two expressions gives

$$g_{k,i} - \hat{g}_{k,i} = \frac{R_{i,+} - R_{i,-}}{2\Delta_k},$$

which proves (3.3). Summing the squared componentwise bounds gives (3.4). $\square$

We now analyze the gradient captured by the Gaussian directions in Algorithm 2.1. Let $\mathcal{F}_k$ contain the algorithmic history before the fresh exploratory directions at iteration k are sampled. Conditionally on $\mathcal{F}_k$,

$$\mathbf{d}_i^{(k)} = \frac{\mathbf{z}_i}{\sqrt{n}}, \quad \mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n), \quad i = 2, \dots, p,$$

independently.

LEMMA 3.3 (Gaussian gradient capture). *Suppose $p \geq 2$, and let $\mathbf{v}_k = \nabla f(\mathbf{x}_k)$. Conditionally on $\mathcal{F}_k$,*

$$(3.5) \quad \sum_{i=2}^p \left(\mathbf{v}_k^\top \mathbf{d}_i^{(k)} \right)^2 \stackrel{d}{=} \frac{\|\mathbf{v}_k\|_2^2}{n} \chi_{p-1}^2.$$

In particular,

$$(3.6) \quad \mathbb{E} \left[\sum_{i=2}^p \left(\mathbf{v}_k^\top \mathbf{d}_i^{(k)} \right)^2 \middle| \mathcal{F}_k \right] = \frac{p-1}{n} \|\mathbf{v}_k\|_2^2.$$

Moreover, for every $q \in (0, 1)$, there exist constants $\alpha > 0$ and $R > 1$, independent of k , such that whenever $\mathbf{v}_k \neq \mathbf{0}$,

$$(3.7) \quad \mathbb{P} \left(\|\hat{\mathbf{g}}_k\|_2 \geq \alpha \|\mathbf{v}_k\|_2, \quad \max_{1 \leq i \leq p} \|\mathbf{d}_i^{(k)}\|_2 \leq R \middle| \mathcal{F}_k \right) \geq q.$$

Proof. For every $i = 2, \dots, p$,

$$\mathbf{v}_k^\top \mathbf{d}_i^{(k)} = \frac{\mathbf{v}_k^\top \mathbf{z}_i}{\sqrt{n}} \sim \mathcal{N} \left(0, \frac{\|\mathbf{v}_k\|_2^2}{n} \right),$$

and these random variables are conditionally independent. Equation (3.5) follows immediately, and taking expectations gives (3.6).

To prove (3.7), fix $q \in (0, 1)$. Since $p \geq 2$, a χ_{p-1}^2 random variable is positive almost surely. We can choose $\tau > 0$ such that $\mathbb{P}(\chi_{p-1}^2 \geq \tau) \geq 1 - \frac{1-q}{2}$. Setting $\alpha = \sqrt{\tau/n}$, equation (3.5) and $\|\hat{\mathbf{g}}_k\|_2^2 \geq \sum_{i=2}^p (\mathbf{v}_k^\top \mathbf{d}_i^{(k)})^2$ imply

$$\mathbb{P}(\|\hat{\mathbf{g}}_k\|_2 \geq \alpha \|\mathbf{v}_k\|_2 \mid \mathcal{F}_k) \geq 1 - \frac{1-q}{2}.$$

Similarly, since each $\|\mathbf{z}_i\|_2/\sqrt{n}$ is almost surely finite and $\|\mathbf{d}_1^{(k)}\|_2 \leq 1$, we can choose $R > 1$ such that $\mathbb{P}(\max_{1 \leq i \leq p} \|\mathbf{d}_i^{(k)}\|_2 \leq R \mid \mathcal{F}_k) \geq 1 - \frac{1-q}{2}$. A union bound then gives (3.7). $\square$

The next proposition bounds the objective decrease produced by the trial step for a general frame.

PROPOSITION 3.4 (Decrease for a general frame). *Suppose Assumption 3.1 holds and $\mathbf{g}_k \neq \mathbf{0}$. Let*

$$\mathbf{s}_k = -\Delta_k \frac{\mathbf{g}_k}{\|\mathbf{g}_k\|_2}, \quad \mathbf{x}_k^+ = \mathbf{x}_k + \mathbf{D}_k \mathbf{s}_k,$$

and define $\beta_k := \|\mathbf{D}_k\|_2$. Then

$$(3.8) \quad f(\mathbf{x}_k) - f(\mathbf{x}_k^+) \geq \Delta_k \|\mathbf{g}_k\|_2 - \frac{L\Delta_k^2}{2} (Q_k + \beta_k^2).$$

Consequently,

$$(3.9) \quad \rho_k \geq 1 - \frac{L\Delta_k(Q_k + \beta_k^2)}{2\|\mathbf{g}_k\|_2}.$$

Proof. Let $\mathbf{h}_k = \mathbf{D}_k \mathbf{s}_k$. Since $\hat{\mathbf{g}}_k = \mathbf{D}_k^\top \nabla f(\mathbf{x}_k)$, we have $\nabla f(\mathbf{x}_k)^\top \mathbf{h}_k = \hat{\mathbf{g}}_k^\top \mathbf{s}_k$. Using $\mathbf{g}_k^\top \mathbf{s}_k = -\Delta_k \|\mathbf{g}_k\|_2$ and Lemma 3.2,

$$\begin{aligned} \nabla f(\mathbf{x}_k)^\top \mathbf{h}_k &= \mathbf{g}_k^\top \mathbf{s}_k + (\hat{\mathbf{g}}_k - \mathbf{g}_k)^\top \mathbf{s}_k \\ &\leq -\Delta_k \|\mathbf{g}_k\|_2 + \|\hat{\mathbf{g}}_k - \mathbf{g}_k\|_2 \|\mathbf{s}_k\|_2 \\ &\leq -\Delta_k \|\mathbf{g}_k\|_2 + \frac{LQ_k}{2} \Delta_k^2. \end{aligned}$$

Also, $\|\mathbf{h}_k\|_2 \leq \|\mathbf{D}_k\|_2 \|\mathbf{s}_k\|_2 = \beta_k \Delta_k$. The descent lemma therefore gives

$$\begin{aligned} f(\mathbf{x}_k^+) &\leq f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^\top \mathbf{h}_k + \frac{L}{2} \|\mathbf{h}_k\|_2^2 \\ &\leq f(\mathbf{x}_k) - \Delta_k \|\mathbf{g}_k\|_2 + \frac{L\Delta_k^2}{2} (Q_k + \beta_k^2), \end{aligned}$$

which proves (3.8). Dividing by the predicted reduction $\Delta_k \|\mathbf{g}_k\|_2$ gives (3.9). $\square$

3.2. Global convergence. We first show that the squared trust-region radii are summable. Let

$$(3.10) \quad \mathcal{S} := \{k : \rho_k \geq \eta \text{ and } \|\mathbf{g}_k\|_2 \geq \kappa \Delta_k\}$$

denote the set of iterations at which the trust-region radius is expanded.

LEMMA 3.5 (Summability of the trust-region radii). *Suppose Assumption 3.1 holds, $\Delta_{\min} = 0$, and the radius is updated according to (3.1). Then*

$$(3.11) \quad \sum_{k=0}^{\infty} \Delta_k^2 < \infty.$$

In particular,

$$(3.12) \quad \lim_{k \rightarrow \infty} \Delta_k = 0.$$

Proof. For any $k \in \mathcal{S}$, the definition of ρ_k and (3.10) give

$$f(\mathbf{x}_k) - f(\mathbf{x}_k^+) = \rho_k \Delta_k \|\mathbf{g}_k\|_2 \geq \eta \kappa \Delta_k^2.$$

Since the right-hand side is strictly positive, the trial point is accepted. Because f is bounded below by a finite constant f_{low} , summing over all successful expansion steps yields

$$\sum_{k \in \mathcal{S}} \Delta_k^2 \leq \frac{f(\mathbf{x}_0) - f_{\text{low}}}{\eta \kappa} < \infty.$$

For every iteration k , the update rule (3.1) implies

$$\Delta_{k+1}^2 \leq \gamma_1^2 \Delta_k^2 + \gamma_2^2 \Delta_k^2 \mathbf{1}_{\{k \in \mathcal{S}\}}.$$

Summing this inequality from $k = 0$ to N yields

$$(1 - \gamma_1^2) \sum_{k=1}^N \Delta_k^2 + \Delta_{N+1}^2 \leq \gamma_1^2 \Delta_0^2 + \gamma_2^2 \sum_{k \in \mathcal{S}} \Delta_k^2.$$

Because $\gamma_1 \in (0, 1)$, we have $1 - \gamma_1^2 > 0$. Taking $N \rightarrow \infty$ establishes (3.11), which directly implies $\lim_{k \rightarrow \infty} \Delta_k = 0$. $\square$

This leads to subsequential convergence of the gradient norm.

THEOREM 3.6 (Subsequential convergence). *Suppose Assumption 3.1 holds and $p \geq 2$. Let the exploratory directions be sampled independently as*

$$\mathbf{d}_i^{(k)} = \frac{\mathbf{z}_i}{\sqrt{n}}, \quad \mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n), \quad i = 2, \dots, p.$$

Then

$$(3.13) \quad \liminf_{k \rightarrow \infty} \|\nabla f(\mathbf{x}_k)\|_2 = 0 \quad \text{almost surely.}$$

Proof. Suppose, to the contrary, that there exist $\varepsilon > 0$ and an index K_0 such that

$$(3.14) \quad \|\nabla f(\mathbf{x}_k)\|_2 \geq \varepsilon, \quad \forall k \geq K_0.$$

Choose a probability threshold

$$(3.15) \quad q > \max \left\{ \frac{1}{2}, \frac{\log(1/\gamma_1)}{\log(\gamma_2/\gamma_1)} \right\}.$$

By Lemma 3.3, there exist constants $\alpha > 0$ and $R > 1$ such that the event

$$G_k := \left\{ \|\hat{\mathbf{g}}_k\|_2 \geq \alpha \|\nabla f(\mathbf{x}_k)\|_2, \quad \max_{1 \leq i \leq p} \|\mathbf{d}_i^{(k)}\|_2 \leq R \right\}$$

satisfies $\mathbb{P}(G_k \mid \mathcal{F}_k) \geq q$. On G_k , we have $Q_k \leq \sqrt{p} R^2$ and $\|\mathbf{D}_k\|_2^2 \leq p R^2$. Applying Lemma 3.2 under (3.14) gives

$$\|\mathbf{g}_k\|_2 \geq \|\hat{\mathbf{g}}_k\|_2 - \|\mathbf{g}_k - \hat{\mathbf{g}}_k\|_2 \geq \alpha \varepsilon - \frac{L\sqrt{p} R^2}{2} \Delta_k.$$

Since $\Delta_k \rightarrow 0$ by Lemma 3.5, for all sufficiently large k we have $\gamma_2 \Delta_k \leq \Delta_{\max}$ and

$$(3.16) \quad \|\mathbf{g}_k\|_2 \geq \frac{\alpha \varepsilon}{2}.$$

Proposition 3.4 then ensures that, on G_k and for all sufficiently large k ,

$$\rho_k \geq \eta \quad \text{and} \quad \|\mathbf{g}_k\|_2 \geq \kappa \Delta_k.$$

Because $\gamma_2 \Delta_k \leq \Delta_{\max}$ holds for all large k , the update rule (3.1) on G_k produces $\Delta_{k+1} = \gamma_2 \Delta_k$ without truncation, while on the remaining iterations $\Delta_{k+1} \geq \gamma_1 \Delta_k$.

Let $I_k := \mathbf{1}_{G_k}$. For all sufficiently large k ,

$$(3.17) \quad \log \Delta_{k+1} - \log \Delta_k \geq I_k \log \gamma_2 + (1 - I_k) \log \gamma_1.$$

Since $\mathbb{E}[I_k \mid \mathcal{F}_k] \geq q$, the strong law of large numbers for bounded martingale differences implies

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{k=K}^{K+N-1} I_k \geq q \quad \text{almost surely.}$$

Averaging (3.17) over N steps therefore yields

$$\liminf_{N \rightarrow \infty} \frac{\log \Delta_{K+N} - \log \Delta_K}{N} \geq q \log \gamma_2 + (1 - q) \log \gamma_1 > 0 \quad \text{almost surely,}$$

where the strict inequality follows from (3.15). This implies $\Delta_k \rightarrow \infty$, contradicting $\Delta_k \rightarrow 0$ from Lemma 3.5. Thus (3.14) cannot hold, establishing (3.13). $\square$

Theorem 3.6 establishes almost-sure global convergence in the subsequential sense, which is the primary convergence guarantee in derivative-free and randomized trust-region methods [8, 6].

To investigate under what conditions the entire sequence of gradient norms converges to zero, fix any $\varepsilon > 0$ and define the index set of large gradients:

$$(3.18) \quad \mathcal{K}_\varepsilon := \{k : \|\nabla f(\mathbf{x}_k)\|_2 > \varepsilon\}.$$

LEMMA 3.7. *Under the assumptions of Theorem 3.6, for every $\varepsilon > 0$,*

$$(3.19) \quad \sum_{k \in \mathcal{K}_\varepsilon \cap G_k} \Delta_k < \infty \quad \text{almost surely.}$$

If, additionally, the renewal property for probabilistically accurate models holds along $\mathcal{K}_\varepsilon$ [2, 9] (that is, the run length of consecutive unsuccessful iterations between occurrences of G_k is stochastically dominated by a geometric distribution), then

$$(3.20) \quad \sum_{k \in \mathcal{K}_\varepsilon} \Delta_k < \infty \quad \text{almost surely.}$$

Proof. Choose the good-frame event G_k as in the proof of Theorem 3.6 with conditional probability $q > 1/2$. Because $\Delta_k \rightarrow 0$, whenever $k \in \mathcal{K}_\varepsilon \cap G_k$ and k is sufficiently large, we have $\|\mathbf{g}_k\|_2 \geq \frac{\alpha\varepsilon}{2}$ and $\rho_k \geq \eta$. Consequently,

$$f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) \geq \frac{\eta\alpha\varepsilon}{2} \Delta_k.$$

Because f is bounded below, summing this decrease over all such iterations establishes (3.19). Under the renewal property, the total sum of radii over $\mathcal{K}_\varepsilon$ is bounded by a constant multiple of the sum over $\mathcal{K}_\varepsilon \cap G_k$, yielding (3.20). $\square$

Recall that $\beta_k = \|\mathbf{D}_k\|_2$. Since the spectral norm is bounded by the Frobenius norm, we have

$$\mathbb{E}[\beta_k^2 \mid \mathcal{F}_k] \leq \mathbb{E}[\|\mathbf{D}_k\|_F^2 \mid \mathcal{F}_k] = \|\mathbf{d}_1^{(k)}\|_2^2 + \sum_{i=2}^p \mathbb{E}\left[\frac{\|\mathbf{z}_i\|_2^2}{n}\right] \leq 1 + (p-1) = p,$$

which immediately implies $\sup_k \mathbb{E}[\beta_k \mid \mathcal{F}_k] \leq \sqrt{p} < \infty$ and $\sup_k \mathbb{E}[\beta_k^2 \mid \mathcal{F}_k] \leq p < \infty$. Together with Lemma 3.5, these moment bounds imply

$$(3.21) \quad \lim_{k \rightarrow \infty} \beta_k \Delta_k = 0 \quad \text{almost surely,}$$

and, whenever (3.20) holds,

$$(3.22) \quad \sum_{k \in \mathcal{K}_\varepsilon} \beta_k \Delta_k < \infty \quad \text{almost surely.}$$

THEOREM 3.8 (Full global convergence). *Under the assumptions of Theorem 3.6, if (3.20) holds for every $\varepsilon > 0$, then*

$$(3.23) \quad \lim_{k \rightarrow \infty} \|\nabla f(\mathbf{x}_k)\|_2 = 0 \quad \text{almost surely.}$$

Proof. Suppose, for contradiction, that (3.23) does not hold. In view of Theorem 3.6, there then exists $\varepsilon > 0$ and infinitely many disjoint pairs of indices $a_j < b_j$ such that

$$\|\nabla f(\mathbf{x}_{a_j})\|_2 \leq \varepsilon, \quad \|\nabla f(\mathbf{x}_{b_j})\|_2 > 2\varepsilon,$$

while

$$\|\nabla f(\mathbf{x}_k)\|_2 > \varepsilon, \quad \forall a_j < k < b_j.$$

By the Lipschitz continuity of the gradient (Assumption 3.1),

$$\varepsilon < \|\nabla f(\mathbf{x}_{b_j})\|_2 - \|\nabla f(\mathbf{x}_{a_j})\|_2 \leq L \sum_{k=a_j}^{b_j-1} \|\mathbf{x}_{k+1} - \mathbf{x}_k\|_2.$$

Since $\|\mathbf{x}_{k+1} - \mathbf{x}_k\|_2 \leq \|\mathbf{D}_k \mathbf{s}_k\|_2 \leq \beta_k \Delta_k$, we can partition the summation as

$$\sum_{k=a_j}^{b_j-1} \|\mathbf{x}_{k+1} - \mathbf{x}_k\|_2 \leq \beta_{a_j} \Delta_{a_j} + \sum_{k=a_j+1}^{b_j-1} \beta_k \Delta_k.$$

As $j \rightarrow \infty$, the first term $\beta_{a_j} \Delta_{a_j} \rightarrow 0$ almost surely by (3.21). Furthermore, all intermediate indices $k \in \{a_j + 1, \dots, b_j - 1\}$ belong to $\mathcal{K}_\varepsilon$. Because $\sum_{k \in \mathcal{K}_\varepsilon} \beta_k \Delta_k < \infty$ almost surely by (3.22), the tail sum $\sum_{k=a_j+1}^{b_j-1} \beta_k \Delta_k$ tends to zero as $j \rightarrow \infty$. Consequently, the entire right-hand side converges to zero, which contradicts the strict lower bound $\varepsilon > 0$.

Therefore, (3.23) holds almost surely. $\square$

Theorem 3.6 (and Theorem 3.8) establishes first-order global convergence for a fixed deterministic objective. In our LLM fine-tuning experiments, minibatches change across iterations, so asymptotic convergence to a stationary point of the population loss is not claimed.

4. Numerical Experiments. We evaluate MpSub on full-parameter fine-tuning of causal language models, comparing against the zeroth-order optimizer MeZO [11] under matched computational budgets. We also examine step-size sensitivity and the effect of the subspace dimension.

4.1. Experimental setup. Experiments are conducted on CommitmentBank (CB) from the SuperGLUE benchmark, a three-class natural language inference task. Following [11], we use 100 training examples, 50 validation examples, and the 56-example validation split as the test set. We evaluate two autoregressive models from the OPT family: OPT-125M ($n \approx 1.25 \times 10^8$ parameters) and OPT-350M ($n \approx 3.5 \times 10^8$ parameters). All parameters are stored and optimized in FP32 with a batch size of 8. Results are averaged over three random seeds, with min-max ranges reported.

Because zeroth-order methods do not compute gradients, computational cost is measured by the total number of forward passes on the training objective. Each method is allotted a budget of 8,400 forward passes. For MpSub with $p = 20$, which takes 200 optimization steps, each iteration requires $2p + 2 = 42$ evaluations. MeZO uses 2 evaluations per step, therefore taking 4,200 steps. Periodic development evaluations are recorded separately and excluded from this budget.

MpSub uses the same fixed hyperparameters across both models: $p = 20$, initial radius $\Delta_0 = 10^{-1}$, contraction factor $\gamma_1 = 0.5$, expansion factor $\gamma_2 = 2.0$, success threshold $\eta = 0.1$, $\Delta_{\min} = 10^{-12}$, and $\Delta_{\max} = 1.0$. For MeZO, the perturbation scale is $\epsilon = 10^{-3}$, and the learning rate η_M is selected by development accuracy over a grid: $\{10^{-8}, 10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}\}$ on OPT-125M and $\{10^{-7}, 10^{-6}, 10^{-5}\}$ on OPT-350M.

4.2. Matched-budget comparison. Table 4.1 reports test and development accuracy across three random seeds under the common budget of 8,400 forward passes.

On OPT-125M, MpSub with the default radius $\Delta_0 = 10^{-1}$ achieves a mean development accuracy of 0.773 and a development loss of 0.674, compared to 0.760 and 0.703 for MeZO, while test accuracies are similar (0.673 versus 0.685). On OPT-350M, MpSub achieves a mean test accuracy of 0.690 and development accuracy of 0.787, compared to 0.685 and 0.780 for MeZO.

Figure 4.1 shows development loss across forward passes. On OPT-125M, MeZO drops quickly initially but then plateaus, whereas MpSub descends steadily to a lower

TABLE 4.1

Matched-budget fine-tuning results on CommitmentBank (8,400 forward passes). Values are means over three random seeds, with min-max ranges in parentheses. The MeZO learning rate is selected by mean development accuracy.

Model	Method	Dev loss	Dev acc.	Test acc.
OPT-125M	MpSub ($\Delta_0 = 10^{-1}$)	0.674 (0.620–0.723)	0.773 (0.740–0.820)	0.673 (0.661–0.679)
	MeZO ($\eta_M = 10^{-6}$)	0.703 (0.625–0.784)	0.760 (0.740–0.800)	0.685 (0.679–0.696)
OPT-350M	MpSub ($\Delta_0 = 10^{-1}$)	0.738 (0.715–0.774)	0.787 (0.760–0.820)	0.690 (0.679–0.714)
	MeZO ($\eta_M = 10^{-6}$)	0.679 (0.582–0.743)	0.780 (0.720–0.880)	0.685 (0.679–0.696)

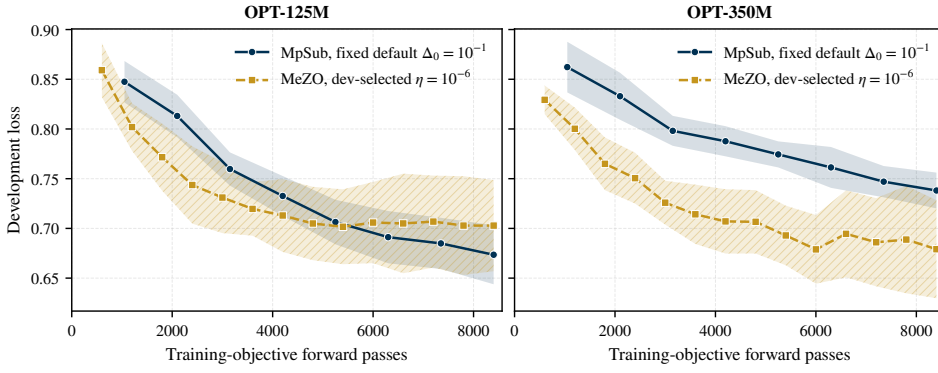


FIG. 4.1. Development loss versus training forward passes for OPT-125M (left) and OPT-350M (right). Shaded bands represent ± 1 standard error of the mean across three random seeds.

final loss (0.674 vs. 0.703). On OPT-350M, MeZO achieves a lower loss during training, but both methods reach similar final test accuracy (0.690 for MpSub vs. 0.685 for MeZO).

4.3. Step-size sensitivity. Figure 4.2 compares the sensitivity of MeZO across learning rates against MpSub across initial radii $\Delta_0 \in \{10^{-3}, 10^{-2}, 3 \times 10^{-2}, 10^{-1}, 3 \times 10^{-1}\}$ on OPT-125M.

MeZO is sensitive to its learning rate: varying η_M by one order of magnitude from 10^{-6} reduces test accuracy by 0.048 at 10^{-7} and by 0.084 at 10^{-5} , requiring a grid search (42,000 forward passes on OPT-125M and 25,200 on OPT-350M) to find an effective rate.

In contrast, MpSub shows little variation across the tested range: mean test accuracy remains between 0.655 and 0.678 across all five initial radii from $\Delta_0 = 10^{-3}$ to 3×10^{-1} . Even when Δ_0 is initialized conservatively at 10^{-3} , successful steps double Δ_k until it reaches a scale where progress can be made. The default choice $\Delta_0 = 10^{-1}$ therefore works across both model sizes without retuning.

4.4. Effect of the subspace dimension. Table 4.2 examines the effect of varying the subspace dimension $p \in \{5, 10, 15, 20, 25, 30\}$ on OPT-125M over 100 iterations.

As p increases from 5 to 20, development accuracy rises from 0.70 to 0.78 and

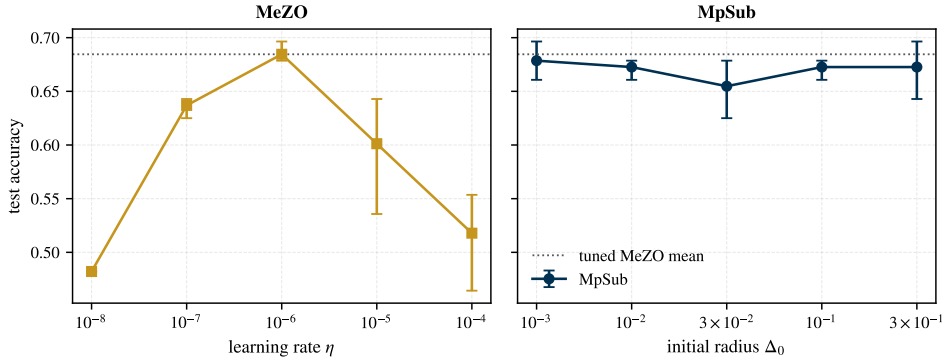


FIG. 4.2. Step-size sensitivity on OPT-125M under 8,400 forward passes. Left: MeZO test accuracy across learning rates η_M . Right: MpSub test accuracy across initial radii Δ_0 .

TABLE 4.2

Ablation on subspace dimension p for OPT-125M (CommitmentBank, 100 steps, $\Delta_0 = 10^{-1}$, single seed). Forward passes per step equal $2p + 2$; run times measured on an NVIDIA RTX 4060 Ti GPU.

p	Forward passes / step	Dev acc.	Test acc.	Time / step (s)
5	12	0.70	0.643	3.0
10	22	0.72	0.661	4.4
15	32	0.74	0.661	6.0
20	42	0.78	0.679	7.7
25	52	0.78	0.679	8.7
30	62	0.76	0.661	10.1

test accuracy from 0.643 to 0.679, consistent with the gradient capture bound in Lemma 3.3. Beyond $p = 20$, accuracy plateaus while evaluation cost and per-step latency continue to increase (from 3.0 seconds at $p = 5$ to 10.1 seconds at $p = 30$). For the budget considered here, $p = 20$ captures most of the performance gain while keeping per-step cost moderate.

5. Conclusion. We proposed MpSub, which combines a random subspace with trust-region step-size control for derivative-free LLM fine-tuning. For smooth deterministic objectives, we proved that $\liminf_{k \rightarrow \infty} \|\nabla f(\mathbf{x}_k)\|_2 = 0$ almost surely under Gaussian direction sampling and the safeguarded radius update, and established full convergence under the renewal property. On CommitmentBank, MpSub achieved results comparable to tuned MeZO on OPT-125M and OPT-350M under the same forward-pass budget, while using the same trust-region settings for both models.

Several questions remain open. The convergence analysis treats a fixed deterministic objective, whereas the LLM experiments resample the minibatch between iterations. Extending the analysis to changing minibatches is therefore the most direct theoretical next step. It would also be useful to study whether curvature information can be introduced into the subspace model without returning to the quadratic interpolation cost that motivated the linear model used here.

While this study evaluates classification on OPT-125M and OPT-350M, testing on larger models, broader task benchmarks, and generative settings is a natural next step. In addition, because the $2p$ directional evaluations within each iteration are

mutually independent, parallel evaluation could substantially reduce wall-clock time. Finally, **MpSub** could be combined with parameter-efficient fine-tuning methods such as LoRA, where the trust-region mechanism would operate over a smaller parameter space.

Acknowledgements. The authors thank the developers of the MeZO and LOZO codebases, on which the experimental infrastructure builds.

REFERENCES

- [1] C. AUDET AND W. HARE, *Derivative-free and blackbox optimization*, Springer, Heidelberg, 2017.
- [2] A. S. BANDEIRA, K. SCHEINBERG, AND L. N. VICENTE, *Convergence properties of trial search direction and trust-region methods based on probabilistic models*, SIAM J. Optim., 24 (2014), pp. 1238–1264, <https://doi.org/10.1137/130919307>.
- [3] C. CARTIS, J. FOWKES, AND Z. SHAO, *Randomised subspace methods for non-convex optimization, with applications to nonlinear least-squares*, arXiv preprint arXiv:2211.09873, (2022).
- [4] C. CARTIS AND L. ROBERTS, *Scalable subspace methods for derivative-free nonlinear least-squares optimization*, Math. Program., 199 (2023), pp. 461–524.
- [5] Y. CHEN, Y. ZHANG, L. CAO, K. YUAN, AND Z. WEN, *Enhancing zeroth-order fine-tuning for language models with low-rank structures*, in International Conference on Learning Representations (ICLR), 2025.
- [6] A. R. CONN, N. I. M. GOULD, AND P. L. TOINT, *Trust Region Methods*, SIAM, Philadelphia, 2000.
- [7] A. R. CONN, K. SCHEINBERG, AND L. N. VICENTE, *Global convergence of general derivative-free trust-region algorithms to first- and second-order critical points*, SIAM J. Optim., 20 (2009), pp. 387–415.
- [8] A. R. CONN, K. SCHEINBERG, AND L. N. VICENTE, *Introduction to Derivative-Free Optimization*, SIAM, Philadelphia, 2009.
- [9] S. GRATTON, C. W. ROYER, L. N. VICENTE, AND Z. ZHANG, *Complexity and global rates of trust-region methods based on probabilistic models*, IMA J. Numer. Anal., 38 (2017), pp. 1579–1597.
- [10] D. KOZAK, S. BECKER, A. DOOSTAN, AND L. TENORIO, *A stochastic subspace approach to gradient-free optimization in high dimensions*, Computational Optimization and Applications, 79 (2021), pp. 339–368, <https://doi.org/10.1007/s10589-021-00271-w>.
- [11] S. MALLADI, T. GAO, E. NICHANI, A. DAMIAN, J. D. LEE, D. CHEN, AND S. ARORA, *Fine-tuning language models with just forward passes*, in Advances in Neural Information Processing Systems, vol. 36, 2023, pp. 53038–53075.
- [12] M. MENICKELLY, *Avoiding geometry improvement in derivative-free model-based methods via randomization*, arXiv preprint arXiv:2305.17336, (2023).
- [13] Y. NESTEROV AND V. SPOKOINY, *Random gradient-free minimization of convex functions*, Foundations of Computational Mathematics, 17 (2017), pp. 527–566.
- [14] P. XIE AND Y. X. YUAN, *A new two-dimensional model-based subspace method for large-scale unconstrained derivative-free optimization: 2D-MoSub*, Optimization Methods and Software, 41 (2025), <https://doi.org/10.1080/10556788.2025.2601670>.
- [15] Z. YU, P. ZHOU, S. WANG, J. LI, M. TIAN, AND H. HUANG, *Zeroth-order fine-tuning of LLMs in random subspaces*, in Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
- [16] Y. YUAN, *Subspace techniques for nonlinear optimization*, in Some Topics in Industrial and Applied Mathematics, R. Jeltsch, D. Q. Li, and I. H. Sloan, eds., Higher Education Press, Beijing, 2007, pp. 206–218.
- [17] Y. YUAN, *A review on subspace methods for nonlinear optimization*, in Proceedings of the International Congress of Mathematicians (ICM 2014), vol. 4, 2014, pp. 807–827.